A PERFECT INNER FACTOR IN A COMPOSITE REGULAR IMMERSION IS ITSELF A REGULAR IMMERSION (NOETHERIAN TARGETS)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18015-18025 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18027-18034, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Used in the chapter's lci factorization-through-affine-space arguments; the two-reference reduction is complete, not a pointer.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Divisors, lemma Noetherian-scheme-regular-ideal (equivalence of regularity notions and local-ring testing in the Noetherian world); Divided Power Algebra, lemma regular-sequence (the local algebra factorization theorem).

Source reference. Original lemma 063I: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18034–18034. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09PW: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18034–18034. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $i : Z \to Y$ and $j : Y \to X$ be immersions of schemes. Assume*

(1) *X is locally Noetherian,*
(2) *$j \circ i$ is a regular immersion, and*
(3) *i is perfect.*

Then i and j are regular immersions.

Proof. Since X (and hence Y) is locally Noetherian all 4 types of regular immersions agree, and moreover we may check whether a morphism is a regular immersion on the level of local rings, see Divisors, Lemma 063I. Thus the result follows from Divided Power Algebra, Lemma 09PW. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.